

An Introduction to Simplicial Homology

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1 Introduction

This document is currently a rough draft write-up of an expedition into algebraic topology. The main text followed was James Munkres' Elements of Algebraic Topology, but auxiliary texts and resources were also used - to be referenced yet. The document is currently incomplete, missing some sections, proofs, and an exercises section (which has content in it but is not linked in the main.tex). We try our best to introduce main topics in a coherent and ordered manner. Much of the auxiliary theory that we have delved into, like category theory, may not be correctly phrased (and this would be mainly my (Abdul's) fault, because I am the one that keeps adding it in). The algebraic topology theory should be sound - as this is what most of the reading time has gone into. Many topics might be rushed right now, that is also my fault. At this stage, we care more about getting down what needs to be in the paper, and polish will come later, unifying the style and pacing.

2 Definitions, Lemmas, and Propositions of Simplices

The simplex generalizes the notion of a triangle from 2-d space to its n-dimensional analogue. More formally, given a geometrically independent set $\{a_0, a_1, \dots, a_n\}$ of vertices, for $N > n$, an n -simplex in $\mathbb{R}^N$ is the set Σ of all $x \in \mathbb{R}^N$ satisfying

$$x = \sum_{i=0}^n a_i t_i, \quad \text{where} \quad \sum_{i=0}^n t_i = 1 \quad \text{and} \quad t_i \geq 0.$$

The t_i 's are called the **Barycentric coordinates** of x , and are uniquely determined by x . The barycentric coordinates t_i , as functions of x with respect to the vertices $\{a_0, \dots, a_n\}$, are continuous functions [See exercises]. A 0-simplex is a vertex $\{a_0\}$, a 1-simplex is the line segment joining a_0 and a_1 , the 2-simplex is the filled-in triangle, the 3-simplex is the filled-in tetrahedron, and so on. A (proper) **face** is the simplex generated by a (proper) subset of the generating vertices.

One could also begin with the notion of the **unit n – simplex** by taking the generating set of vertices to be $\left\{\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}\right\} \subseteq \mathbb{R}^{n+1}$. Then, we define the affine n -simplex Φ to be the composition of an affine transformation with the simplex, i.e. $\Phi = T \circ \Sigma$. Furthermore, we can impose an orientation on this by taking equivalence classes with respect to even and odd permutations of the vertices. This will be useful in the future when discussing the homology groups of simplicial complexes. Note that this way of constructing simplices guarantees geometric independence.

Denoting the n -simplex by $\sigma \subseteq \mathbb{R}^N$, we also have the following properties:

1. (1) σ is compact, and is exactly equal to the intersection of all (not necessarily compact) convex sets $U \subset \mathbb{R}^N$ containing the vertices $\{a_0, \dots, a_n\}$.
2. (2) let s be the opposite face of a_0 . Then σ is equal to the union of all rays connecting points of s to a_0 ; two such distinct rays intersect only at a_0 .

The barycentric coordinates of $x \in \sigma \subseteq \mathbb{R}^N$ are positive, for a non-zero-dimensional simplex σ if and only if x is an interior point of σ . The point at which all of the barycentric coordinates are positive is called the *barycenter* of σ .

2.1 Simplicial Complexes

Take a collection of a few different simplices embedded in some higher dimensional space, and "glue" them together by matching vertices to vertices, edges to edges, faces to faces, etc. This is a simplicial complex. Even if no two distinct simplices share a common face, that is still a valid complex.

Formally, if K is a simplicial complex, then $\forall \sigma \in K$, every face of σ is also in K , and if two simplices intersect, their intersection is a face of both of them. Δ^N shall denote the complex formed by only the n -simplex. It is the "simplest" simplicial complex.

Let $K \subset \mathbb{R}^N$ be a simplicial complex and let $x \in K$. Since K is the union of simplices, the point x has relative barycentric coordinates based on the simplices that it lies within. Notice that x could lie on a proper face of some higher dimensional simplex, but would thus necessarily have zeroes in its barycentric coordinates.

2.2 Abstract Simplicial Complexes

An abstract simplicial complex is the combinatorial set construction of a simplicial complex. Formally, an abstract simplicial complex K has two associated sets: its **vertex** set V_K and its **simplices** S_K , such that every singleton $\{v\}$ in V_K is an element of S_K (note, we mean $\{v\} \in S_K$), and $\forall \sigma \in S_K$, if $\sigma' \subset \sigma$, then $\sigma' \in S_K$. It is often convenient to talk about the *maximal* simplices in S_K (simplices that are not contained in a higher dimensional simplex, and have largest dimension in the complex – rewrite this def), since simplices by definition contain all of their faces.

Every simplex is a simplicial complex (complex for short). Take

$$\begin{aligned} V_K &= \{a, b, c\}, \\ S_K &= \{\{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\} \end{aligned}$$

Indeed, all singletons of V_K are in S_K , and for any simplex in S_K , we notice that all of its subsets are also in S_K . We exclude $\emptyset$ for obvious reasons. This particular simplicial complex is indeed a 2-simplex. If we were to remove the set $\{a, b, c\}$ from S_K , we would obtain the hollow triangle. This is actually the only set we are allowed to remove from this particular complex, as otherwise simplices in S_K would be missing some of their faces. $\{a, b, c\}$ is also our maximal simplex.

A complex whose maximal simplices have dimension k will be said to have dimension k . This idea will be generalized further later. For $p < k$, the p -skeleton of a complex, denoted K^p , is the union of all of its p -dimensional faces. For example, K^0 denotes its vertices, K^1 includes its vertices and its edges, K^2 includes its 2-d faces, edges, and vertices, and so on. K^k would result in the original complex.

2.2.1 Geometric Realizations

It would be convenient to translate between abstract simplices and the simplices we defined above as subsets of euclidean space. In category theoretic terms, we define a functor $|-| : \mathbf{ASC} \rightarrow \mathbf{Top}$, where $\mathbf{ASC}$ is the category of abstract simplicial complexes, whose objects are complexes and whose morphisms are set functions between their vertex sets. One such functor $|-| : \mathbf{ASC} \rightarrow \mathbf{Top}$ may act on objects as follows: We may assert a bijection $\varphi : V_K \rightarrow \mathfrak{D}$ for a subset $\mathfrak{D} \subset \mathbb{N}$. Then for the canonical unit vectors ε_k , define:

$$|K| = \bigcup_{\sigma \in S_K} \text{conv}\{\varepsilon_{\varphi(s)}\}_{s \in \sigma}.$$

Effectively, we label the vertices, and take the convex hulls of all the faces of the simplices. One can verify that this indeed defines a functor between the categories. Later, we define complexes of arbitrary dimension. For these, one may need to use generalized euclidean space, or a Hilbert space like $\ell_2(V_K)$ in order to achieve a faithful realization or embedding.

If K is an abstract simplicial complex, then we denote its geometric realization (also know as its *polytope*, or its *underlying space*) by $|K|$. From now on, the convention is that K denotes abstract complexes, and $|K|$ is a subset of some (possibly generalized) euclidean space $\mathbb{E}^J$.

If we start with $|K|$ and describe it combinatorically, we recover the original abstract simplicial complex. "Abstractification" is also a functor $A : \mathbf{GSC} \rightarrow \mathbf{ASC}$, where $\mathbf{GSC}$ is the category of geometric simplicial complexes, whose morphisms are simplicial maps, which will be defined soon, and are effectively the induced topological map between spaces given by a mapping between vertex sets. If we retranslate it using the functor defined above, we only recover a complex that is homeomorphic to the original one (homeomorphisms will be defined below. Essentially, there is an affine transformation between them). In the diagram below, A is a forgetful functor that forgets geometric data:

$$\mathbf{GSC} \xrightarrow{A} \mathbf{ASC} \xrightarrow{|-|} \mathbf{Top}$$

2.2.2 Polytope Topology

The *polytope topology* of a complex $|K|$ is defined via a unifying concept called the "final topology", which is in general the topology of every direct

limit in the category of topological spaces. In particular, polytope topology is the coherent topology, which uses several inclusion maps from each simplex σ to K . Openness in $|K|$ with respect to the polytope topology is defined as the openness of the preimages of each of these inclusion maps with respect to normal subspace topology. In other words, $U \subset |K|$ is open in $|K|$ if and only if $U \cap \sigma$ is open in σ for all $\sigma \in K$ (an abuse of notation; we examine openness as if σ were a topological subset of $|K|$). Equivalently, a set V is closed in $|K|$ if and only if $V \cap \sigma$ is closed for every σ .

It is obvious to see that if $U \subset |K|$ is subspace open, then it is open in the polytope topology as well. If K is locally finite (that is, every simplex σ only intersects finitely-many other simplices τ), then the notions are equivalent: if U intersects n simplices, then for $x \in U$, we can take $\varepsilon = \min\{\varepsilon_1, \dots, \varepsilon_n\}$ for which $U \cap B_{\varepsilon_i}(x) \subset \sigma_i$ to obtain $U \cap B_\varepsilon(x) \subset \sigma_i$. Thus $U \cap \sigma_i$ is either empty (trivially open) or is non-empty and open, thus U is open in the polytope topology.

A counter-example for a non-locally finite space is the complex K made up of vertices at $(0, 1)$ and $(n, 0)$ for each $n \in \mathbb{Z} \setminus \{0\}$, with edges from each $(n, 0)$ to $(0, 1)$. Since one can take the union

$$U = \bigcup_{n \in \mathbb{Z} \setminus \{0\}} (B_{\frac{1}{|n|}}((0, 1)) \cap \sigma_n)$$

where σ_n is the edge connecting $(0, 1)$ and $(n, 0)$. By construction, U is a subset created from intersecting open sets with each simplex in K , and thus is open in the polytope topology. Since there is no upper bound on n , it is impossible to fit an epsilon ball inside of U that lies in every simplex. Therefore no open set A exists in $\mathbb{R}^N$ such that $A \cap |K| = U$, i.e., U is not open in the subspace topology.

2.2.3 Continuous Functions on Simplices

Let $f : |K| \rightarrow X$ be a function. We have the necessary and sufficient condition that f is continuous if and only if $f|_\sigma$ is continuous for every $\sigma \in K$. The sufficiency is obvious since σ is a subspace of $|K|$. If $f|_\sigma$ is continuous for each $\sigma \in K$, then for a closed subset $C \subset X$, $f^{-1}(C) \cap \sigma = f|_\sigma^{-1}(C)$ is closed in $|K|$ by continuity of $f|_\sigma$.

2.3 Homeomorphisms Between Simplicial Complexes

Suppose K and L are simplicial complexes. Let $f : K^0 \rightarrow L^0$ be a set function between the vertices of K and the vertices of L . f is a morphism of **ASC**. Suppose that if the vertices $v_1, v_2, \dots, v_n$ span a simplex of K , then $f(v_1), f(v_2), \dots, f(v_n)$ span a simplex of L . Then f extends to a continuous function $g : |K| \rightarrow |L|$ such that

$$x = \sum_{i=0}^n t_i a_i \implies g(x) = \sum_{i=0}^n t_i f(a_i).$$

Proof. Suppose $v_1, v_2, \dots, v_n$ spans a simplex $\sigma \in K$ and suppose f is as above, such that $f(\{v_1, v_2, \dots, v_n\})$ spans a simplex $\tau \in L$. We collect like terms. If some $v_i = v_j$, we collect $t_{ij}f(v_i) = (t_i + t_j)f(v_i)$ so that it is clear that the t_k 's in $\sum t_i f(v_i)$ still sum up to 1, despite the fact that the vertices under f may span a lower dimensional simplex (which they would if f is not injective, like the case described). Since the barycentric coordinates are continuous, the linear combination is continuous from $\sigma \rightarrow \tau$. Hence $g|_\sigma$ is continuous for every $\sigma \in K \iff g : |K| \rightarrow |L|$ is continuous. $\square$

g is called the linear simplicial map induced by f . The composition of simplicial maps $|K| \rightarrow |L|$ and $|L| \rightarrow |M|$ is also a simplicial map (this is easy to verify).

Lemma 2.0.1. *Suppose that $f : K^0 \rightarrow L^0$ was bijective. Then the simplicial map $g : |K| \rightarrow |L|$ is a homeomorphism of complexes.*

g is then called a simplicial homeomorphism or isomorphism.

Proof. Both f and f^{-1} induce simplicial maps g and h respectively. The composition of these maps is a simplicial map. By definition,

$$h \circ g(x) = h\left(\sum_{i=0}^n t_i f(v_i)\right) = \sum_{i=0}^n t_i f^{-1} \circ f(v_i) = \sum_{i=0}^n t_i v_i = x,$$

and thus $h = g^{-1}$ is a continuous inverse of g . $\square$

Analogous to finite graphs being isomorphic to a subgraph of K_n for some n , all finite simplicial complexes K are isomorphic to a subcomplex of Δ^N for some N , by the lemma. Simply assert any set bijection between K^0 and $(\Delta^N)^0$.

2.4 Infinite-Dimensional Simplices

We do not need to restrict ourselves to N -complexes (complexes whose maximal simplices have dimension N). Let J be an arbitrary index set, and let $\mathbb{R}^J$ be arbitrary space. Take a subspace $\mathbb{E}^J \subset \mathbb{R}^J$ such that $(x_\alpha)_{\alpha \in J}$ has only finitely many non-zero coordinates. One can check that it is indeed a subspace of the vector space $\mathbb{R}^J$. Under the topology induced by the metric $d(x, y) = \max_{\alpha \in J} |x_\alpha - y_\alpha|$, $\mathbb{E}^J$ is called *generalized euclidean space*. If K is a complex in $\mathbb{E}^J$, then the dimension of K is the largest dimension of a simplex in K and is denoted $\dim K$. If there is no such maximal-dimension simplex, then K is said to have *infinite dimension*.

All of our previous should work for arbitrary simplices. The reader can verify this (or check the exercises because we will likely be the one to verify this).

2.5 Pasting Maps and Quotient Constructions

am yet to write: connected sum thing, edge labeling, the full subcomplex condition, dimension-collapse preventing,

This section formalizes constructions like the torus that identifies the opposite edges together. It requires a surjective map f from one vertex set to a "labeling" set (which becomes our quotient space once we realize it geometrically). Given a finite complex L , we take $f : L^0 \rightarrow K^0$ to be surjective, where $|K|$ is the geometric realization of an abstract complex S corresponding to our L^0 , i.e. whose vertices are of the form $\{f(v_1), f(v_2), \dots, f(v_n)\}$. Then $g : |L| \rightarrow |K|$ is a simplicial map that is surjective. Since $|L|$ is compact (L^0 is finite by assumption $\iff |L|$ is compact) and $|K|$ is Hausdorff, g is a closed map, and thus is also a quotient map. g is called the *associated pasting map*.

3 Homology

It can be very difficult to show that two spaces are not homeomorphic, because it requires proving that no homeomorphism exists between them. Instead, it is often easier to find a difference between invariants of spaces. Many spaces that might seem "difficult" to analyze, like a torus, can be analyzed very easily using the simple geometric objects that we have thus defined. In

fact, since isomorphisms are functorial from abstract complexes to their topological polytopes (though, polytope to abstract complex is not functorial), we can define the homology groups of a space, which is a very nice invariant that depends only on the underlying space yet is easily described by simplices of the complex, and the triangulation of the space does not matter.

3.1 Free Abelian Groups

to be filled in: abelian groups with basis,

generating free abelian groups from a set S

finitely-generated and the finiteness condition for arbitrary bases.

direct sums (in particular how it works with subgroups),

fundamental theorem of finitely-generated abelian groups

In order to talk about homology groups, we must first briefly discuss free abelian groups. In short, an abelian group G is free if it has a basis; i.e., if $\{g_\alpha\}_{\alpha \in \mathbb{J}}$ is a basis, then every $g \in G$ can be uniquely written as a finite formal linear combination

$$g = \sum_{i=1}^n n_i g_i, \quad \text{where } n_i \in \mathbb{Z} \text{ are integers.}$$

Of course, we can take coefficients in any ring R , but it is sufficient to take $\mathbb{Z}$. If this representation is not unique (i.e., relations between the generators exist so they are not linearly independent), then G is said to be only *generated* by $\{g_\alpha\}_{\alpha \in \mathbb{J}}$. If $\mathbb{J}$ is finite, then G is called *finitely generated*. A free abelian group does not require that $\mathbb{J}$ is finite, only that every element can be written using finitely-many terms. The rank of a finitely-generated free abelian group is the number of basis elements it has, and this is invariant for any choice of basis of G . In general, any two bases of G have the same cardinality.

Any set S can generate a free abelian group G : Let G be the collection of all functions $\phi : S \rightarrow \mathbb{Z}$ such that for all but finitely-many $x \in S$, $\phi(x) \neq 0$. Define the addition of two elements of G to be the integer addition of their values. The set of characteristic functions $\{\phi_x\}_{x \in S}$ is a basis of G . Indeed, if $\phi \in G$, then

$$\phi = \sum_{x \in S} n_x \phi_x,$$

where $\phi(x) = n_x$. Of course, this is always possible because G contains every such function ϕ , and because they are only nonzero for finitely-many $x \in S$.

By identifying every characteristic function with its element, we can thus represent every element $g \in G$ by the formal sum $g = \sum n_\alpha g_\alpha$, where $g_\alpha \in S$ corresponds to ϕ_{g_α} .

The torsion subgroup of any group G is the set of all elements $\{g \in G \mid g^n = 0, \text{ for } n \in \mathbb{N}\} = T \leq G$. The torsion subgroup vanishes if G is free; the converse is not necessarily true.

3.1.1 Direct Sums

Let G be a group, and let $\{G_\alpha\}_{\alpha \in \mathbb{J}}$ be a collection of subgroups. If every $g \in G$ can be written uniquely as a finite sum $\sum g_\alpha$ for $g_\alpha \in G_\alpha$, then G is called the internal direct sum of the subgroups G_α , i.e.,

$$G = \bigoplus_{\alpha \in \mathbb{J}} G_\alpha$$

. If each g can be written as $\sum g_\alpha$ but not uniquely, then G is called the sum of the groups G_α . The sum is direct if and only if $\bigcap G_\alpha = \{0\}$. In general, we have that G is a free abelian group if and only if G is the direct sum of $\{G_\alpha\}$, where each $G_\alpha = \langle g_\alpha \rangle$ is the infinite cyclic group generated by g_α .

A *direct summand* of G is a subgroup G_1 such that there exists a subgroup G_2 where $G = G_1 \oplus G_2$. If H_1 and H_2 are subgroups of G_1 and G_2 respectively, then $H_1 \oplus H_2$ is direct, and moreover,

$$\frac{G}{H_1 \oplus H_2} \cong \frac{G_1}{H_1} \oplus \frac{G_2}{H_2}.$$

. In particular, we have that if $G = G_1 \oplus G_2$, then $G/G_1 \cong G_2$. The converse is not true; take $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/n$ as a counterexample.

3.1.2 Fundamental Theorem of Abelian Groups

Theorem 3.1. *If G is a free abelian group of rank n and H is a subgroup of G , then H is free of rank $r \leq n$, and moreover, there is a basis $e_1, \dots, e_n$ of G such that there are integers $t_1, \dots, t_k > 1$, where:*

1. $t_1 e_1, t_2 e_2, \dots, t_k e_k, e_{k+1}, \dots, e_n$ is a basis of H
2. $t_i \mid t_{i+1}$ for each i .

The numbers $t_1, \dots, t_k$ are uniquely determined by G and H , but the basis $e_1, \dots, e_n$ is not.

From this theorem, we also get the following:

Theorem 3.2 (The Fundamental Theorem of Finitely-Generated Abelian Groups). *Let G be a finitely-generated abelian group and let T be its torsion subgroup.*

1. *There exists a free abelian subgroup $H \leq G$ of rank β such that $G = H \oplus T$.*
2. *There are finite cyclic subgroups $T_1, \dots, T_k$ of order $t_1, \dots, t_k > 1$ respectively, such that $t_i \mid t_{i+1}$ and $T = T_1 \oplus \dots \oplus T_k$.*
3. *The numbers $t_1, \dots, t_k$ and β are uniquely determined by G*

β is called the *beti number* of G . The numbers $t_1, \dots, t_k$ are called the torsion coefficients of G . The rank of the subgroups H and T_k are uniquely determined, but the subgroups themselves are not.

Proof. trivial □

Thus, any finitely-generated abelian group can be written canonically as

$$G \cong \mathbb{Z} \oplus \dots \oplus \mathbb{Z} \oplus \mathbb{Z}/t_1 \oplus \dots \oplus \mathbb{Z}/t_k$$

where each t_i divides t_{i+1} .

3.2 p-Chains

fill in: C_p is free abelian generated from the collection of p -simplices in K
orientation
i think this is all we need

3.2.1 Orientation

In order to define chains, we must first define the orientation of a simplex. Let $v_0 v_1 \dots v_n$ denote the n -simplex with vertices $v_0, \dots, v_n$. Denote the oriented simplex by $[v_0 v_1 \dots v_n]$, where we take equivalence classes based on even or odd permutations of the vertices. We note that if σ_1 and σ_2 denote the same

simplex with opposite orientation, then we say $[\sigma_1] = -[\sigma_2]$. If the context is clear, we drop the brackets.

A p -chain c of a complex K is formally a function on the set of oriented p -simplices $c : \{\text{oriented } p\text{-simplices}\} \rightarrow \mathbb{Z}$ such that

1. if $\sigma = -\sigma'$ then $c(\sigma) = -c(\sigma')$
2. $c(\sigma) = 0$ for all but finitely-many $\sigma \in C_p(K)$

We add p -chains by adding their values. The collection of p -chains forms a free abelian group denoted $C_p(K)$. We assert that this group is trivial for $p < 0$ and $p > \dim K$. For a p -simplex $\sigma \in K$, we can also identify it with its unique p -chain c_σ where $c_\sigma(\sigma) = 1$, $c_\sigma(\sigma') = -1$ if σ' has the opposite orientation of the same simplex, and $c_\sigma(\tau) = 0$ for all other p -simplices τ . We refer to such a chain as an *elementary p -chain*.

By orienting (arbitrarily) every p -simplex in K , we can use them as a basis to generate $C_p(K)$. The orientation is not unique and choices do need to be made.

p -chains should not be thought of geometrically, though they can be occasionally (e.g., the counterclockwise boundary chain of a triangle). They are thought of better as an algebraic bookkeeping mechanism.

3.3 Boundary Maps

We now define a homomorphism between the chain groups of a complex K . Let $\sigma = [v_0, v_1, \dots, v_p]$ be an oriented p -simplex with $p > 0$. Define the boundary operator as follows: let $\partial_p : C_p(K) \rightarrow C_{p-1}(K)$ such that

$$\partial_p(\sigma) = \bigoplus_{j=0}^p (-1)^j [v_0, v_1, \dots, \hat{v}_j, \dots, v_p]$$

where the hat indicates the omittance of v_j .

For example, the boundary of an elementary 1-chain is $\partial_1([v_1, v_2]) = v_2 \ominus v_1$. For a triangle $\sigma = [v_0, v_1, v_2]$, the formula gives us

$$\partial_2(\sigma) = [v_1, v_2] \ominus [v_0, v_2] \oplus [v_0, v_1].$$

If you were to draw this, you would find that the topological boundary of the simplex is what you would expect it to be, oriented counter-clockwise. From

now on, we drop the ' $\oplus$ ' notation for the '+' notation. In truth, while it is a formal sum, the chain is really a function into $\mathbb{Z}$, so the latter notation is okay.

This is well-defined in that $\partial_p(-\sigma) = -\partial_p(\sigma)$. Another very important property of the boundary operator is that $\partial_p \circ \partial_{p+1} = 0$. We shall show this directly, but first, we define two important terms:

$\sigma \in C_p(K)$ is called a *cycle* if $\partial_p(\sigma) = 0$. If $\exists d \in C_{p+1}(K)$ such that $\partial_{p+1}(d) = \sigma$, then we say that σ is a *boundary*, or that σ *bounds*. It is clear from the statement that all boundaries are cycles. If the converse were also true, then this entire expedition would have all been a massive waste of time.

Lemma 3.2.1. $\partial_p \circ \partial_{p+1} = 0$.

Proof. Let $\sigma = [v_0, v_1, \dots, v_{p+1}] \in C_{p+1}(K)$ be an oriented $(p+1)$ -chain of K .

$$\begin{aligned}
\partial_p \circ \partial_{p+1}(\sigma) &= \bigoplus_{j=0}^{p+1} (-1)^j \partial_p([v_0, \dots, \hat{v}_j, \dots, v_{p+1}]) \\
&= \bigoplus_{j=1}^{p+1} (-1)^j \left(\bigoplus_{i=0}^{j-1} (-1)^i [v_0, \dots, \hat{v}_i, \dots, \hat{v}_j, \dots, v_{p+1}] \right) \\
&\quad \oplus \bigoplus_{j=0}^p (-1)^j \left(\bigoplus_{i=j+1}^{p+1} (-1)^{i-1} [v_0, \dots, \hat{v}_j, \dots, \hat{v}_i, \dots, v_{p+1}] \right) \\
&= \bigoplus_{j=1}^{p+1} \left(\bigoplus_{i=0}^{j-1} (-1)^{i+j} [v_0, \dots, \hat{v}_i, \dots, \hat{v}_j, \dots, v_{p+1}] \right) \\
&\quad \ominus \bigoplus_{i=1}^{p+1} \left(\bigoplus_{j=0}^{i-1} (-1)^{i+j} [v_0, \dots, \hat{v}_j, \dots, \hat{v}_i, \dots, v_{p+1}] \right) \\
&= 0
\end{aligned}$$

where the last difference is achieved via relabelling the i, j indices. $\square$

Homology is really the study of generalized holes in a space. For example, a 1-cycle may or may not bound a 2-dimensional subcomplex. We are interested in cases where such a chain is non-trivial (i.e. it does not bound), as this indicates a hole in the complex. Under homeomorphism, these holes are invariant.

We define $Z_p(K) := \ker \partial_p \leq C_p(K)$ and call them the p -cycles of K , and similarly $B_p(K) := \operatorname{im} \partial_{p+1} \leq C_p(K)$ called the p -boundaries of K . We note that $B_p(K) \leq Z_p(K)$ by the preceding lemma.

Let $p > 0$. We define the p -th homology group as follows:

$$H_p(K) := Z_p(K)/B_p(K) = \ker \partial_p / \operatorname{im} \partial_{p+1}$$

We postpone the $p = 0$ case for later.

If K is a finite complex, then $Z_p(K)$ is finitely-generated, implying that $H_p(K)$ is finitely-generated. By the fundamental theorem of finitely-generated groups,

$$H_p(K) \cong \mathbb{Z}^\beta \oplus \mathbb{Z}/q_1 \oplus \cdots \oplus \mathbb{Z}/q_m$$

β is called the *betti number*, and $q_1, \dots, q_m$ are called the *torsion coefficients* of K in dimension p . These are, in fact, topological invariants of the underlying space $|K|$.

3.4 Homology Examples

Example 3.2.1. Let K be as pictured in 3.1, that is, made up of the 1-simplices e_1, e_2, e_3, e_4 arranged counterclock-wise in a square. Since there are no 2-simplices in the complex, $Z_2(K) = B_2(K) = \{0\} \implies H_2(K) = \{0\}$. The general 1-chain on K has the form $c = \sum n_j e_j$, so computing $\partial(c) = (n_{j+1} - n_j)v_j$ indicates that the value of each vertex v_j in the boundary is $(n_{j+1} - n_j)$ where we simply take n_5 to just mean n_1 . Thus, c is a cycle if and only if $n_{j+1} = n_j$ for each j , including $n_4 = n_1$. Obviously this forces all coefficients to be equal which indeed matches our intuition. Thus, every 1-cycle is of the form $c = n(e_1 + e_2 + e_3 + e_4)$, implying that $Z_1(K) = \langle (e_1 + e_2 + e_3 + e_4) \rangle \cong \mathbb{Z}$. Since $C_2(K) = \{0\}$, $B_1(K) = \operatorname{im} \partial_2 = \{0\}$. Thus,

$$H_1(K) = Z_1(K)/B_1(K) \cong \mathbb{Z}$$

An identical argument applies to complexes formed from the 1-simplices $[v_0, v_1], \dots, [v_{n-1}, v_n], [v_n, v_0]$ and their vertices.

Example 3.2.2. Let L be pictured as in 3.2, now with the edge e_5 going through it and the oriented 2-simplices σ_1 and σ_2 . We see that a general 2-chain on L has the form $d = m_1\sigma_1 + m_2\sigma_2$, and that $\partial(d) = m_1(e_1 + e_2 - e_5) + m_2(e_3 + e_4 + e_5)$, which is 0 if and only if $m_1 = m_2 = 0$. Thus,

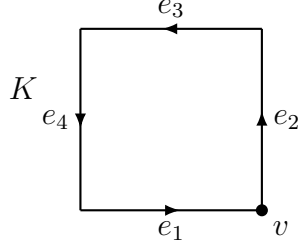


Figure 3.1:

$Z_2(L) = \{0\} \implies H_2(L) = \{0\}$. Now, by calculating the boundary of a general 1-chain $c = \sum n_j e_j$, we obtain that c is a cycle if and only if

$$\begin{aligned}
0 &= \partial(c) \\
&= \sum n_k \partial(e_k) \\
&= n_1(v_2 - v_1) + n_2(v_3 - v_2) + n_3(v_4 - v_3) + n_4(v_1 - v_4) + n_5(v_3 - v_1) \\
&= (n_4 - n_5 - n_1)v_1 + (n_1 - n_2)v_2 + (n_2 + n_5 - n_3)v_3 + (n_3 - n_4)v_4 \\
&\iff n_1 = n_2, n_3 = n_4, n_4 - n_5 - n_1 = n_3 - n_5 - n_2 = -(n_2 + n_5 - n_3) = 0
\end{aligned}$$

Therefore, we can freely choose n_1 and n_3 , and the other coefficients are determined. Choosing $n_1 = 0, n_3 = 1$ gives a basis element $c_1 = e_3 + e_4 + e_5$, and choosing $n_1 = 1, n_3 = 0$ gives the second basis element $c_2 = e_1 + e_2 - e_5$. Thus, $Z_1(L) = \langle c_1, c_2 \rangle \cong \mathbb{Z}^2$. Since these basis elements happen to be exactly the boundaries of σ_1 and σ_2 , we thus have $\langle \partial(\sigma_1), \partial(\sigma_2) \rangle \subset B_1(L)$ which implies $Z_1(L) = B_1(L) \implies H_1(L) = \{0\}$. Our intuition is that H_1 collects data about holes in the space - since there are no holes, the group is trivial.

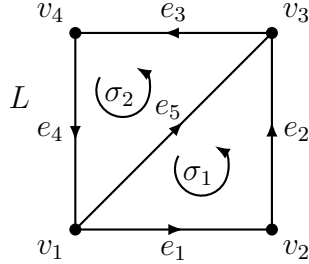


Figure 3.2:

Eventually, it will become clear that the homology groups of K rely only on the underlying space $|K|$. This works out well for us: isomorphisms between abstract simplicial complexes, which are the objects we use to build our p -chain groups $C_p(K)$, induce homeomorphisms between topological space. Geometric realizations are indeed functorial: in particular, $K \cong L \implies |K| \cong |L|$. That means we can build these invariants for relatively complicated spaces, which we will see in the next section, using relatively simple objects. The converse is not true in general: homeomorphisms of spaces do not imply isomorphic triangulations of the same space - the choice of triangulation is not in general functorial. This may sound unfortunate, but from another perspective, it means that the triangulation of a topological space does not affect its homology groups. We don't have to make any choices that affect our homology groups.

Let us define some more terminology in order to deal with some slightly more complicated complexes. Let K be a complex and L a subcomplex. We say that a p -chain $c \in C_p(K)$ is *carried* by L if $c(\sigma) = 0$ for every p -simplex $\sigma \in K \setminus L$. We shall also say that two p -chains c and c' are *homologous* if $c = c' + \partial d$ for a $(p+1)$ -chain d . We will write $c \sim c'$, and it is clear that this is an equivalence relation. In particular, c is homologous to 0 if $c = \partial d$, or more commonly, c *bounds*.

We proceed with a few more of the aforementioned complicated examples. The strategy we employ can be essentially summed up by trivializing all of the possible cycles by making them homologous to each other. Classifying all cycles in this way allows us to easily deal with them later without computing all of Z_p .

Example 3.2.3. Consider the complex M . Intuitively, we can see that all of its homology groups are trivial. But proving this directly is an absolute pain, because of how many choices of cycles we can consider. Instead, we will consider some arbitrary chain, and show that it is homologous to a chain carried by the boundary and some "protruding" edges in a way that the only cycles that exist are carried entirely by the boundary (see 3.4). In doing so, we show that such a chain is a cycle if and only if it is homologous to a cycle carried by the boundary. Indeed, if c and c' are homologous, then $\partial c = \partial c' + \partial \circ \partial d = \partial c'$, that is, their boundaries are the same. In a sense, we can envision the following procedure to be like "pushing" the chain onto the boundary by a special construction. After that, we only need to check if a cycle carried by the boundary is a boundary of a 2-simplex itself. As for

the procedure itself:

We take an arbitrary 1-chain $c \in C_1(M)$ which has the form $\sum n_i e_i$. Suppose that $c(e_1) = m_1$. We notice that $\partial\sigma_1(e_1) = -1$. So, if we create a new chain $c_1 = c + \partial(m_1\sigma_1)$, then clearly $c \sim c_1$, but more importantly, $c_1(e_1) = 0$. Continuing, we let $m_2 = c_1(e_2)$, and let $c_2 = c_1 + \partial(m_2\sigma_2)$, so that $c_2(e_2) = 0$ (note the coefficient of e_2 in ∂s_1). Since the edge e_1 is not part of the boundary of σ_2 , we still have that $c_2(e_1) = 0 + 0 = 0$. Since $c_2 - c_1 = \partial\sigma_2$ and $c_1 - c = \partial\sigma_1$, we have that $c_2 - c = \partial(\sigma_1 + \sigma_2)$, i.e. $c_2 \sim c$. (this statement proves transitivity of $\sim$ in general, which was left out above). One finds that the same can be done to create a chain $c_3 = c_2 + \partial(m_3\sigma_3)$ where $m_3 = c_2(e_3)$. What we have essentially shown is that every single 1-chain in M is homologous to a 1-chain carried by the subcomplex pictured in 3.4.

It is obvious now that if c is a cycle, then c is homologous to a cycle d carried by the boundary. By previous arguments, d must be of the form $d = n(e_5 + e_6 + e_7 + e_8)$. However, this is exactly the boundary $\partial(n(\sigma_1 + \sigma_2 + \sigma_3 + \sigma_4))$, so that $d \sim 0$. By transitivity, $c \sim 0 \implies Z_1(M) = B_1(M)$, or equivalently, $H_1(M) = \{0\}$.

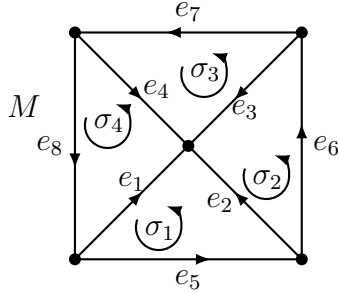


Figure 3.3:

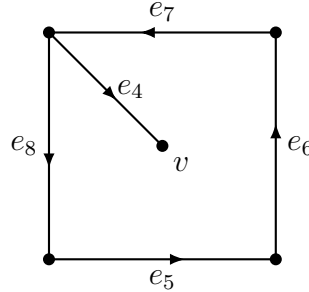


Figure 3.4:

3.4.1 Infinite Homology Groups

Let K be a complex. The group of *infinite p -chains* $C_p^\infty(K)$ is comprised of normal chains c , except that we do not require that $c(\sigma) = 0$ for all but finitely-many p -simplices $\sigma \in K$. In general, this group is not free (though, importantly, it is still abelian, so that quotients are well-defined).

If K is locally finite, then we define a boundary operator

$$\partial_p^\infty : C_p^\infty(K) \rightarrow C_{p-1}^\infty(K)$$

that satisfies Lemma 3.2.1. Indeed, this lemma relies on the fact that the sum can be partitioned into two sums that cancel each other, where $i < j$ and where $j < i$. K being locally finite means that an open ball of $\mathbb{E}^J$ intersects only finitely-many simplices $\sigma \in K$. Thus, a simplex is only a face of finitely many simplices of K . Therefore, the coefficients of the double sum are finite, and thus they cancel, leaving that $\partial_p^\infty \circ \partial_{p+1}^\infty = 0$.

Since this guarantees that $B_p^\infty(K) \subset Z_p^\infty(K)$, we can define the *homology groups of infinite chains* by

$$H_p^\infty(K) := \ker \partial_p^\infty / \operatorname{im} \partial_{p+1}^\infty$$

.

3.5 Homology of a Torus and Other Surfaces

just follow the theorem. go through with the klein bottle, projective space, the $\#$ operator, the classification of particular compact spaces and their homologies. perhaps, mention the triangulation of ≤ 3 dim compact manifolds.

Before we compute the homology groups of these space, we first prove a very useful lemma to make our lives easier.

Lemma 3.2.2. *Let K be the complex whose underlying space $|K|$ is a rectangle (this is an important condition) as pictured in Figure 3.5. Then every cycle is homologous to a cycle carried by the subcomplex $\operatorname{Bd} K$. Orient all of the 2-simplices counter-clockwise and orient the edges arbitrarily. Then:*

1. *Every cycle is homologous to a cycle carried by $\operatorname{Bd} K$.*
2. *if the boundary of a 2-chain is carried by $\operatorname{Bd} K$, then it is a multiple of $\sum \sigma_i$.*

Proof. To prove (1), we essentially implement the exact same algorithm described in Example 3.2.3. Starting with an arbitrary chain, we start by "pushing" it away from the center (i.e. constructing a homologous chain that has a non-zero coefficient for all but at most one edge in the center of the rectangle), and continuing the same procedure for any edge that connects another nonboundary edge to the boundary. We effectively end up with a chain carried by a subcomplex K_0 of K that is exactly a spanning tree of K if you were to removed exactly one edge from the boundary [see Figure 3.6].

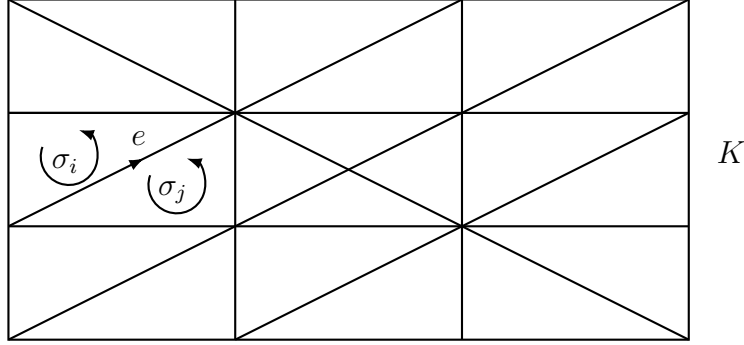


Figure 3.5:

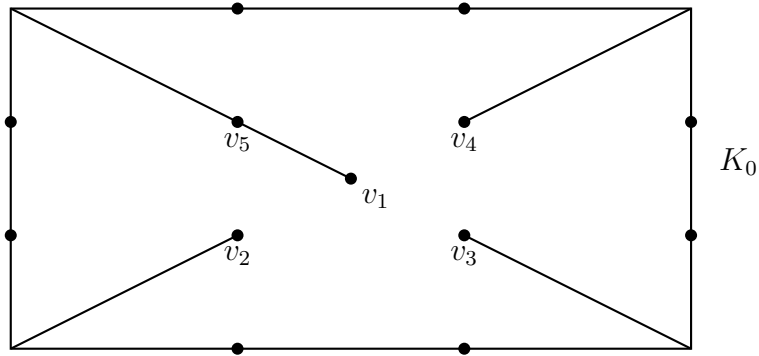


Figure 3.6:

At this point, it is obvious that any 1-cycle is homologous to a 1-cycle carried by $\text{Bd } K$. To prove (2), we let $\gamma = \sum_{\nu} \sigma_{\nu}$. Notice that if σ_i shares an edge with σ_j , then by our chosen orientation, the edges cancel each other out. Thus, $\partial\gamma$ is exactly $\text{Bd } K$. If some σ_k were missing from the sum, then its boundary edges would have nonzero coefficients in the sum. Therefore, if d is a 2-chain such that $\partial d \subset \text{Bd } K$, then $d = n\gamma$. $\square$

This lemma is very useful because the same rectangle can be used to triangulate different 2-surfaces which we will need to classify every compact 2-surface [CHECK THIS PHRASING]. By labeling the vertices in different ways, we can construct the torus T , the Klein bottle, and the projective space $\mathbb{P}^2$.

Theorem 3.3. *Let L be the complex of the rectangle in Figure 3.7 and T denote its image under labeling whose underlying space is a torus. In the fol-*

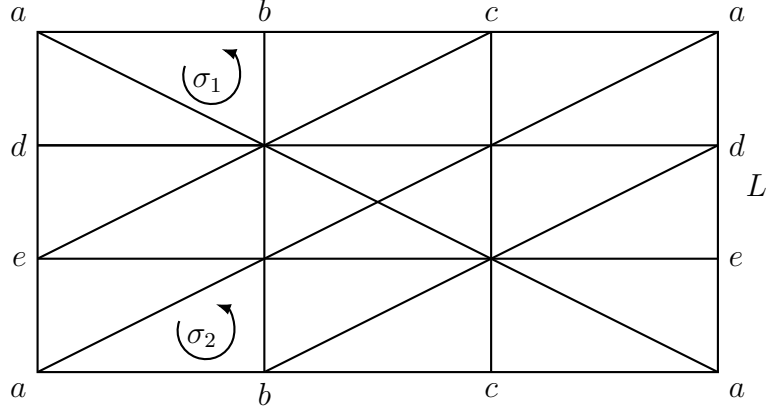


Figure 3.7:

lowing diagrams, we will call the unlabeled rectangle L , but name its labeling as the space we are considering (in this case T). Then $H_1(T) \cong \mathbb{Z} \oplus \mathbb{Z}$, and $H_2(T) \cong \mathbb{Z}$.

Proof. As in Lemma 3.2.3, orient the 2-simplices of the complex counter-clockwise, and let γ denote their sum. Let

$$\begin{aligned} w &= [a, b] + [b, c] + [c, a] \\ z &= [a, d] + [d, e] + [e, a] \end{aligned}$$

Then we show that $\{w, z\}$ is a basis for $H_1(T)$, and γ is a basis for $H_2(T)$, which will prove the claim.

Let $g : |L| \rightarrow |T|$ be the associated pasting map of the labeling, and let $A = g(|\text{Bd } L|)$. By the Lemma, every cycle is homologous to a cycle carried by $\text{Bd } T$. It is clear that under the identification, that such a cycle is of the form $nw + mz$. It is clear that w and z are independent; for if $nw + mz = 0$, then $m = n = 0$ as they share no common edges. This cycle does not bound; for $\partial d = q\partial\gamma$ by the Lemma, but notice that every edge on the boundary now shares an edge with a 2-simplex of opposite orientation under the identification, and thus $\partial d = 0$. Thus, $\partial d = nw + mz = 0$ implies $n = m = 0$, i.e. every 1-cycle that bounds is trivial, so $H_1(T) \cong \langle w, z \rangle \cong \mathbb{Z} \oplus \mathbb{Z}$. This works because the only simplices altered by the labeling are those on the boundary. Moreover, since $\partial\gamma = 0$ and $B_2(T)$ is trivial, $H_2(T)$ is generated by γ , and thus $H_2(T) \cong \mathbb{Z}$. $\square$

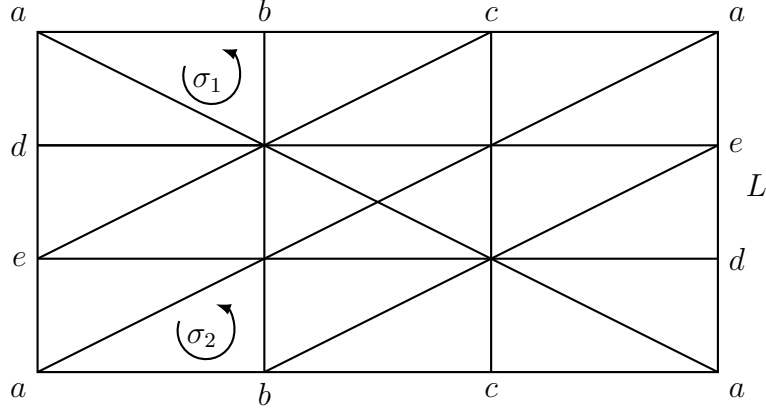


Figure 3.8:

Theorem 3.4. *Let L denote the complex outlined by the rectangle in Figure 3.8, and let S be the image of a morphism under labeling, whose underlying space is called the Klein Bottle. Orient every 2-simplex counter-clockwise and let γ denote their sum. Then*

$$H_1(S) \cong \mathbb{Z} \oplus \mathbb{Z}/2, \quad H_2(S) \cong \{0\}$$

Let

$$\begin{aligned} w &= [a, b] + [b, c] + [c, a] \\ z &= [a, d] + [d, e] + [e, a] \end{aligned}$$

Then the torsion element is represented by the chain z , and the free subgroup of $H_1(S)$ is generated by w .

Proof. Again, let $g : |L| \rightarrow |S|$ be the associated pasting map of the labeling and let $A = g(|\text{Bd } L|)$. By the Lemma, every cycle is homologous to a cycle of the form $nw + mz$. However, this time $\partial\gamma$ is not 0; for the 2-simplices whose boundaries involve the edges of z , their coefficients are not zero; both sides contribute +1 to the boundary, and hence $\partial\gamma = 2z$. Therefore if ∂d is carried by A , then $\partial d = 2qz$ for some integer q . Thus, $2qz = \partial d = nw + mz$ bounds if and only if $n = 0$ and m is even, hence $H_1(S) \cong \mathbb{Z} \oplus \mathbb{Z}/2$. Since $d = q\gamma$ is never a cycle whenever $d \neq 0$, as $\partial d = 2qz$, we have that $H_2(S) = \{0\}$. $\square$

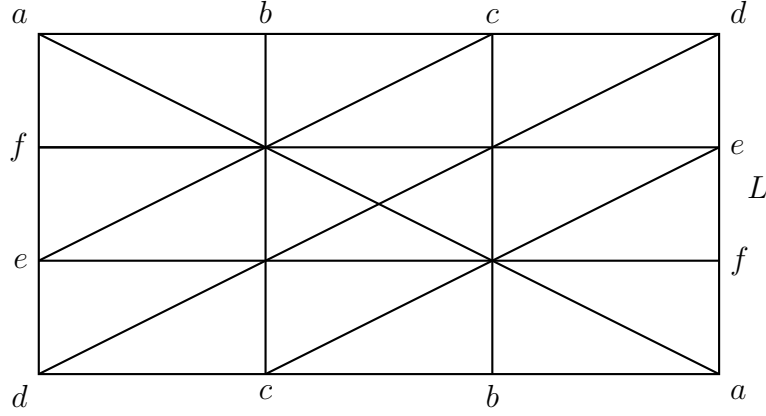


Figure 3.9:

Theorem 3.5. *Let $\mathbb{P}^2$ be the labeled complex denoted in Figure 3.9. Its underlying space is called the projective plane. Then*

$$H_1(\mathbb{P}^2) = \mathbb{Z}/2, \quad H_2(\mathbb{P}^2) = \{0\}$$

Proof. As before, we let $g : |L| \rightarrow |\mathbb{P}^2|$ be the associated pasting map and let $A = g(\text{Bd } L)$, so that A is homeomorphic to a circle. Let γ be the sum of the oriented 2-simplices as before. Let

$$w = [a, b] + [b, c] + [c, d] + [d, e] + [e, f] + [f, a]$$

It is clear that every 1-cycle is homologous to nw , and $\partial d = q\partial\gamma = -2qw$ by examining the coefficients of $\partial\gamma$. Thus, $\partial d = -2qw = nw$ implies that nw bounds if and only if n is even. We conclude that $H_1(\mathbb{P}^2) = \mathbb{Z}/2$. Since no nontrivial d is a cycle, $H_2(\mathbb{P}^2) = \{0\}$. $\square$

3.6 Connected Sums

We can define the *connected sum* of two surfaces S_1 and S_2 by cutting a small hole and gluing together the edges of those holes. Formally, let $D_i \subset S_i$ be open disks for $i = 1, 2$ and take any homeomorphism $h : \partial D_1 \rightarrow \partial D_2$. Define $S_i^\circ = S_i \setminus D_i$, and define $S_1 \# S_2 := (S_1^\circ \cup S_2^\circ) / (x \sim h(x))$. Our choice of homeomorphism doesn't matter here; the spaces derived from two different

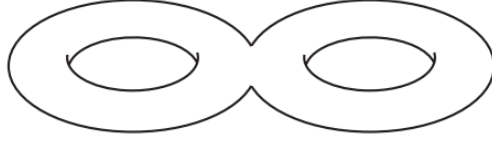


Figure 3.10:

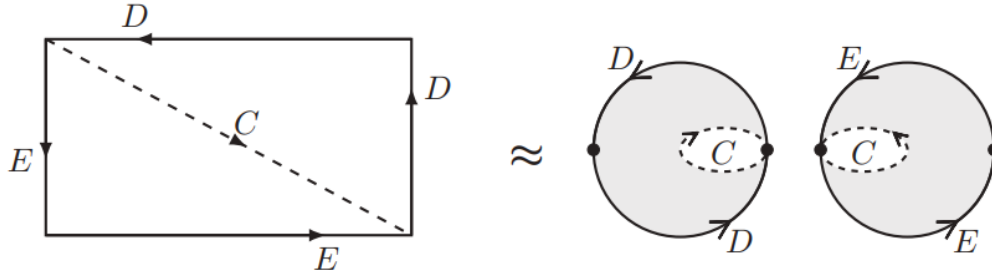


Figure 3.11:

homeomorphisms are themselves homeomorphic. One could simply compose the first homeomorphism with the inverse of the second. For an example of $T \# T$, see Figure 3.10.

3.6.1 Homology of $\mathbb{P}^2 \# \mathbb{P}^2$

One can represent $\mathbb{P}^2 \# \mathbb{P}^2$ via the quotient given in Figure 3.11.

Theorem 3.6. $H_1(\mathbb{P}^2 \# \mathbb{P}^2) \cong \mathbb{Z} \oplus \mathbb{Z}/2$ $H_2(\mathbb{P}^2 \# \mathbb{P}^2) \cong \{0\}$.

Proof. We label L as we did before, and we let z be the top edge (chain) running left to right, and w be the bottom edge running right to left. It is clear as before that any 1-cycle is homologous to a cycle of the form $nw + mz$, but also, $\partial\gamma = 2z + 2w$. From this, it is clear that no nontrivial 2-chain bounds, so that $H_2(\mathbb{P}^2 \# \mathbb{P}^2) = \{0\}$.

Since a 1-chain bounds if and only if it is equal to $\partial\gamma = 2(z+w)$, we can be clever and choose a basis $\{z, z+w\}$ which is indeed still linearly independent. The advantage is that it is now obvious that $nz + m(z+w)$ bounds if and only if $n = 0$ and m is even, by the above, so that $H_1(\mathbb{P}^2 \# \mathbb{P}^2) \cong \mathbb{Z} \oplus \mathbb{Z}/2$.

It is no coincidence that the homology groups are the same as the Klein Bottle, as the two spaces are isomorphic (see Figure 3.12). However in gen-

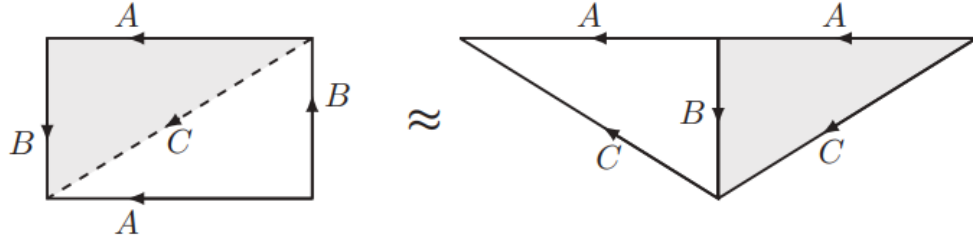


Figure 3.12:

eral, one cannot use one invariant alone to determine the space (unless it is a really good invariant). $\square$

Theorem 3.7. *Every compact surface is homeomorphic to either a sphere, n connected tori, or m connected projective spaces.*

Proof. yeah no. $\square$

3.7 0-Dimensional Homology

augmentation map as a homomorphism

0-dim reduced homology and relation to normal homology - very short section. do the exercises in this section to show any epimorphism works.

Now let us define and deal with 0-dimensional homology groups. Since $C_{-1}(K)$ is by definition trivial, $\ker \partial_0 = C_0(K)$.

step 1: define each C_v set

step 2: prove that they are disjoint and that all vertices by star method are homologous.

step 3: this means the eq classes form a basis for H_0 .

step 4: profit.

Our plan is essentially to show that a basis is a collection of a single vertex from each connected component of $|K|$. If v and w are vertices, then $v \sim w$ if and only if they are the boundary of a 1-chain e , that is, e has the form $a_0 a_1 a_2 \dots a_n$ where $a_0 = v$ and $a_n = w$. Define C_v as follows:

$$C_v := \{\text{St } v' \mid v \sim v'\}$$

We prove that each distinct C_v is exactly the connected components of $|K|$. Clearly C_v is path-connected, since each $\text{St } v'$ is path-connected and $\sim$ is

transitive. Also, we clearly have that $C_v = C_{v'} \iff v \sim v'$. That is, if $C_v \cap C_{v'} = \emptyset$, then each C_v lies in a different connected component of $|K|$. Indeed, since each connected component is a union of simplices which are convex, they are path-connected. If $x \in |K|_\alpha$ was an element in the connected component $|K|_\alpha$, then x lies inside some face of a simplex so that there is a path $[x, v_\alpha]$ that lies completely in $|K|_\alpha$, that also lies completely within $\text{St } v_\alpha$, and hence the connected components lie completely within the geometric realizations of each C_{v_α} .

Let $\{v_\alpha\}$ be a collection with one vertex from each connected component of $|K|$. Given an arbitrary 0-chain c , each of its vertices that lie in the same connected component are homologous to some vertex v_α . Thus, $c \sim \sum n_\alpha v_\alpha$, i.e. c is homologous to a linear combination of the vertices $\{v_\alpha\}$. Since $\partial d = \sum n_\alpha d_\alpha$ is a linear combination of 1-chains from each connected component, we have that $\partial d = c$ implies $\sum \partial d_\alpha = \sum n_\alpha v_\alpha$. This is 0 if and only if each $n_\alpha = 0$. Thus no nontrivial chain bounds.

This effectively proves that $H_0(K) = \mathbb{Z}^r$ where $|K|$ has r connected components. This is okay and all, but one would like to have a simpler version where a connected complex ends up with a trivial homology. This motivates the next topic.

3.7.1 Reduced Homology

We now define the augmentation homomorphism $\varepsilon : C_0 \rightarrow \mathbb{Z}$, which sends the 0-chain $\sum n_\alpha v_\alpha$ to $\sum n_\alpha$. That is, $\varepsilon(v) = 1$ for some vertex v .

We can then define the reduced homology group in dimension 0 to be $\tilde{H}_0(K) := \ker \varepsilon / \text{im } \partial_1$. For all other p , we posit that $\tilde{H}_p(K) = H_p(K)$.

A 0-chain c is homologous to some $c' = \sum n_\alpha v_\alpha$, which bounds only if $n_\alpha = 0$. If $\varepsilon(c) = 0$ then $\varepsilon(c) = \varepsilon(c') = \sum n_\alpha = 0$, which implies that exactly one vertex v_β is solved by the others, i.e. the cosets of $v_\alpha - v_\beta$ form a basis for $\tilde{H}_0(K)$. If $|K|$ has one connected component, then the reduced homology in dimension 0 is trivial, and it holds that $\tilde{H}_0(K) \oplus \mathbb{Z} \cong \mathbb{Z} \cong H_0(K)$. Otherwise, it is free abelian of rank $r - 1$ for r connected components, and once again we have that $\tilde{H}_0(K) \oplus \mathbb{Z} \cong H_0(K)$.

3.8 cones

show its a complex

show examples

It should be kind of obvious intuitively that the reduced homology groups of an n -simplex are trivial, but it requires some work to make formal. We introduce the useful notion of a cone for this.

Effectively, we find some new point w such that any ray emanating from w intersects with $|K|$ in at most one point, and then take the collection of all $(p+1)$ -simplices $wv_0v_1v_2\cdots v_p$ and their faces, where $v_0v_1\cdots v_p$ is a p -simplex of a complex K in $\mathbb{E}^J$. The new space is denoted $w * K$, and one can show that $w * K$ is indeed a complex. K is called the *base* of $w * K$.

The set $\{w, v_0, v_1, \dots, v_p\}$ is indeed geometrically independent for every simplex $v_1\cdots v_p$. If it were not, then the linear combination $rw + \sum_{k=0}^p r_kv_k = 0$ would have more than one non-zero solution (by definition of geometric dependence), and so the intersection of the ray with the simplex would necessarily contain at least two points $(r_{j,1})$ and $(r_{j,2})$, which our choice of w excludes. In fact, its affine hull would contain the entire line segment from w to any point in the simplex.

Since a ray from w to any simplex σ in K intersects at most one point, the intersection between the interior of two simplices with w as a vertex must be empty, for we have that

$$\text{int}(wv_1\cdots v_p) = \bigcup_{u \in \text{int}(v_1\cdots v_p)} (w, u)$$

which implies that a nonempty intersection between two simplices in $w * K$ only occurs when u lies in the interior of two simplices of K , which is impossible as K is a complex. Hence, $w * K$ is a complex.

Note that two cones $z * K$ and $w * K$ are isomorphic via the map induced by sending $z \mapsto w$. Also, it is useful to note that we can always consider that a complex $K \in \mathbb{R}^n$ is embedded into $\mathbb{R}^n \times \{0\}$, so we can always find some geometrically independent point in the higher space in order to make cones.

Example 3.7.1. If we denote Δ^N as the simplicial complex for the simplex $v_0v_1\cdots v_N$ and its faces, then it is clear that Δ^N is isomorphic to $v_0 * \Delta^{N-1}$, where Δ^{N-1} is the complex containing the simplex $v_1\cdots v_N$ and its faces. Therefore every simplex is a cone.

Example 3.7.2. Consider the complex K in $\mathbb{R}^2$, pictured in 3.13, consisting of the vertices $(n, 0)$ for $n \in \mathbb{Z}$ and edges $[n, n+1]$. Taking any other point w with a non-zero y -component gives a geometrically independent point. Observe that $|K|$ is a topological subspace of $\mathbb{R}^2$, but $|w * \mathbb{R}^2|$ is not (that

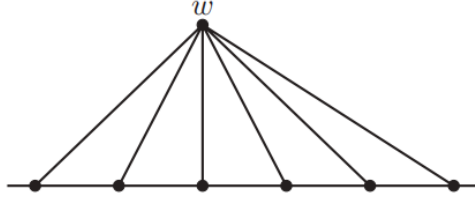


Figure 3.13:

is, the subspace topology does not agree with the polytope topology since $|w * K|$ is not locally finite).

Indeed, when K is locally finite, we can take a minimal index m for which an open ball centered around any vertex z intersecting v_m has smallest distance δ_m . Thus we can take a ball with radius $\delta_m/2$ in the topology of $\mathbb{R}^2$ that is a subset of the ball in the polytope topology. Since the polytope topology is finer than the subspace topology (a trivial fact), we thus attain equality of bases and thus equality of topologies. When we do not have local finite-ness, this fails because we could always take a sequence of balls with radius $\frac{1}{n}$ corresponding to a countably-infinite sequence of simplices, and we will never find some ε -ball in $\mathbb{R}^2$ that is contained inside of this ball. The polytope topology is strictly finer than the subspace topology.

Theorem 3.8. *Let U be a bounded, open, convex set in $\mathbb{R}^n$, and let $w \in U$. If K is a finite complex such that $|K| = \bar{U} \setminus U$, then $w * U$ is a finite complex such that $\bar{U} = |w * K|$.*

Proof. Since U is convex, $|K|$ intersects rays from w at exactly one point, and $\bar{U}$ is exactly the union of all rays emanating from w to points in $|K|$. $\square$

Definition 3.8.1. Let $w_1 * K$ and $w_2 * K$ be cones whose polytopes intersect exactly at $|K|$. Then $(w_1 * K) \cup (w_2 * K)$ is also a complex, and is called the *suspension* of K , denoted $S(K)$. See 3.14

3.9 homology of a cone

Definition 3.8.2. Let $w * K$ be a cone and let $\sigma = [a_0, a_1, \dots, a_p]$ be an oriented p -simplex in K , and let $[w, \sigma]$ denote the oriented simplex $[w, a_0, a_1, \dots, a_p]$. Since interchanging two vertices in the array for σ also corresponds to interchanging two vertices in the array denoted by $[w, \sigma]$, orientation is still well-defined.

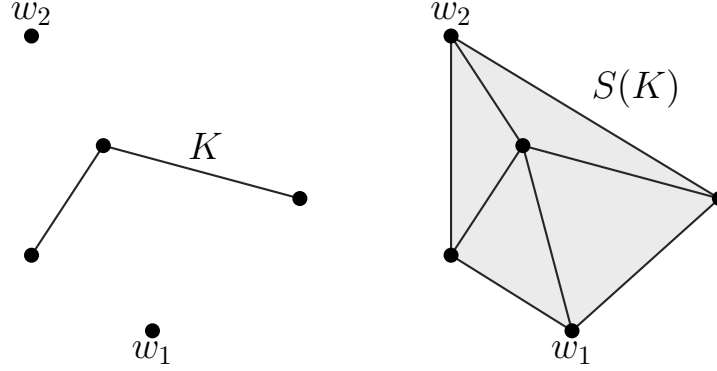


Figure 3.14:

In general, for a p -chain $c = \sum n_i \sigma_i$,

$$[w, \cdot] : C_p(K) \rightarrow C_{p+1}(K) \quad \sum n_i \sigma_i \mapsto \sum n_i [w, \sigma_i]$$

is a homomorphism of chain groups.

The boundary of $[w, \sigma]$ for a simplex σ is computed to be $\sigma - w$ if $\dim \sigma = 0$, or otherwise $\sigma - [w, \partial \sigma]$. The $\dim \sigma = 0$ case is obvious to see. For $\dim \sigma > 0$, let $\sigma = [a_1, \dots, a_{p+1}]$ (this is a convenient labeling), thus:

$$\partial[w, \sigma] = \sigma - \sum_{i=1}^{p+1} (-1)^i [w, a_1, \dots, \hat{a}_i, \dots, a_{p+1}] = \sigma - [w, \partial \sigma].$$

This gives us the general equations for a p -chain c_p :

$$\partial[w, c_p] = \begin{cases} c_p - \varepsilon(c_p)w & \text{if } p = 0 \\ c_p - [w, \partial c_p] & \text{if } p > 0 \end{cases}$$

Noting that the bracket $[w, \cdot]$ being a homomorphism means that it distributes over the sum of simplices in the p -chain, this is not too difficult to see.

Lemma 3.8.1. $w * K$ is path-connected.

Proof. for any $x, y \in |w * K|$, there are simplices $\sigma, \tau \in w * K$ for which $x \in \sigma$ and $y \in \tau$. Without loss of generality, we can assume that σ is a simplex

of the form $wa_0a_1 \dots a_p$ for some simplex $a_0a_1 \dots a_p \in K$, and we can also assume that τ is a simplex of the form $b_0b_1 \dots b_q$ that lies entirely in K . It is clear by convexity that the interval $[x, w]$ lies entirely in $|\sigma|$, and that there exists a simplex $\tau' = wb_0b_1 \dots b_q$ of $w * K$, and the interval $[w, y]$ lies entirely in τ' . Thus there exists an explicit path that traces out $[x, w] \cup [w, y]$. Notice that $|K|$ need not be connected. $\square$

We can now move onto our main theorem for cones:

Theorem 3.9. *Let $w * K$ be a cone. $\forall p$:*

$$\tilde{H}_p(w * K) \cong \{0\}.$$

In general, a complex for which every reduced homology group vanishes is called asyclic.

Proof. Since $w * K$ is connected, $\tilde{H}_0(w * K) \cong 0$. Let $p > 0$, and let z be a p -cycle. We can write $z = c_p + [w, d_{p-1}]$ for some chains c_p, d_{p-1} carried by K . We claim that $z_p = \partial[w, c_p]$:

$$\begin{aligned} z_p - \partial[w, c_p] &= c_p + [w, d_{p-1}] - \partial[w, c_p] \\ &= c_p + [w, d_{p-1}] + c_p - [w, \partial c_p] \\ &= [w, d_{p-1} - \partial c_p], \end{aligned}$$

since $[w, \cdot]$ is a homomorphism. Letting $e_{p-1} = d_{p-1} - \partial c_p$ and taking the boundary on both sides, we get

$$0 = \partial z_p - \partial^2[w, c_p] = \begin{cases} e_{p-1} - \varepsilon(e_{p-1})w & \text{if } p = 1 \\ e_{p-1} - [w, \partial e_{p-1}] & \text{if } p > 1 \end{cases}$$

Since e_{p-1} is the only part that lies entirely in K , we must have that $e_{p-1} = 0$ (as w is geometrically independent to the simplices of e_{p-1}). In particular, $z_p - \partial[w, c_p] = [w, e_{p-1}] = 0$, which proves the claim. $\square$

Theorem 3.10. *Let σ be an n -simplex, and let K_σ be the complex consisting of σ and its faces. If $n > 0$, let Σ^{n-1} be the complex whose polytope is $|\text{Bd } \sigma|$. Orient σ . Then $\tilde{H}_i(\Sigma^{n-1}) \cong 0$ for all $i \neq n - 1$, and $\tilde{H}_{n-1}(\Sigma^{n-1})$ is infinite cyclic and generated by $\partial\sigma$.*

Proof. Since K_σ is a cone, it is acyclic by the theorem. In dimension $i \neq n$, the chain groups are identical:

$$\begin{array}{ccccccccc}
C_n(K_\sigma) & \xrightarrow{\partial_n} & C_{n-1}(K_\sigma) & \xrightarrow{\partial_{n-1}} & \cdots & \xrightarrow{\partial_1} & C_0(K_\sigma) & \xrightarrow{\varepsilon} & \mathbb{Z}. \\
& & \parallel & & \parallel & & \parallel & & \parallel \\
0 & \xrightarrow{\partial'_n} & C_{n-1}(\Sigma^{n-1}) & \xrightarrow{\partial'_{n-1}} & \cdots & \xrightarrow{\partial'_1} & C_0(\Sigma^{n-1}) & \xrightarrow{\varepsilon'} & \mathbb{Z}.
\end{array}$$

Thus it is clear that for $i \neq n-1$, $\tilde{H}_i(K_\sigma) \cong \tilde{H}_i(\Sigma^{n-1}) \cong 0$. Let $n > 1$, then $C_n(\Sigma^{n-1}) \cong 0$ so there can be no boundaries. Thus,

$$\begin{aligned}
H_{n-1}(\Sigma^{n-1}) &\cong Z_{n-1}(\Sigma^{n-1}) = \ker \partial'_{n-1} \\
&= \ker \partial_{n-1} = \text{im } \partial_n \quad \text{as } H_{n-1}(K_\sigma) \cong 0 \\
&= \langle \sigma \rangle \cong \mathbb{Z}, \quad \text{as } C_{n-1}(K_\sigma) \text{ has no torsion.}
\end{aligned}$$

The case for $n = 1$ is identical, just replacing ∂_{n-1} with ε instead. $\square$

Since Σ^{n-1} is convex, it is homeomorphic to the sphere $\mathbb{S}^{n-1}$. Thus we have also computed the homology of the sphere as a corollary.

bracket hom, isom theorem

4 Exercises

Chapter 1: Homology Groups of a Simplicial Complex

Exercises for Section 1.2: Simplices

1. Verify Lemma 1.2.6(1)–(3).

Proof. Big boy proof □

2. Verify Lemma 1.2.6(4) as follows:

- (a) Show that if $x \in \sigma$ and $x \neq a_0, a_1, \dots, a_n$ then x lies in some open line segment contained in σ . [Assume $n > 0$.]

Proof. WRITE YOUR PROOF HERE. □

- (b) Show that a_0 lies in no open line segment contained in σ , by showing that if $a_0 = tx + (1 - t)y$, where $x, y \in \sigma$ and $0 < t < 1$, then $x = y = a_0$.

Proof. WRITE YOUR PROOF HERE. □

3. Verify Lemma 1.2.6(5)

Proof. lolz □

4. Generalize Lemma 1.2.6(2) as follows: Let σ be spanned by $a_0, a_1, \dots, a_n$. Let s be the face of σ spanned by $a_0, a_1, \dots, a_p$ [where $p < n$], and let t be the face opposite s .

- (a) Show that σ is the union of all line segments joining points of s to points of t , and two of these line segments intersect in at most a common end point.

Proof. hihi □

- (b) Show that $\text{Int } \sigma$ is the union of all open line segments joining points of $\text{Int } s$ to points of $\text{Int } t$.

Proof. hehe

□

5. Let U be a bounded open set in $\mathbb{R}^n$. Suppose that U is star-convex relative to the origin: (a) Show that a ray from 0 may intersect $\text{Bd } U$ in more than one point. (b) Show by example that U need not be homeomorphic to Bn

Exercises for Section 1.3: Simplicial Complexes and Simplicial Maps

1. hey (hi) Let K be a simplicial complex; let $\sigma \in K$. When is $\text{Int } \sigma$ open in $|K|$? When is σ open in $|K|$?

Solution 4.0.1. solution instead of a proof shall we!

Yasss okay (im lazy don wanna typeset)

Basically im lowk pretty sure that the interior of simplex is not what we think it is only when its a single vertex, but also like when its lower 'dimensional' (like in simplex terms, like the 'k' in k-simplex is maximal) because then its relative to the space rightt. like basically it needs to not be contained in a bigger simplex and we're good.

like lowk its kind of the same other than thatt. oohhhhhh but wait, open is gonna be relative to the topology of K which is finer than the subspace topology... hmamamm ok im thinking "outloud" (idk what this would be called) ok no distractiions what changes here, finer topology means more things can be open than the subspace top so itll be stuff thats open normally and moree. butttt because we're only looking at σ in K , it doesnt really matter does it because the closure and openness is only relative to σ so we can really just take K as the ambient space and σ as a subspace and we're like lowk back like so back right. sooo to answer, uh hh i think its what it was above right lemme think... alr i thoght about it and it seems right.

for the second part, uh hh this literally can only happen if K is not ocn-nected because thats the only way a closed simplex can also be open (its an iff thing right?). and that σ needs to be its own component.

2. Show that in general, $\text{St } v$ and $\bar{\text{St}} v$ are path connected.

Solution 4.0.2. TRIVIALLLLLLLLL!!!!11!11!!!1 (ill typeset it later walahiiii)

Basically its like easy right, we have a couple way to do it probably. One way is that connected and open iff path connected (i thinkk), or the other way is just that literally just observe that every simplex with the v guy in the middle has an edge to v . assuming 2 different points u and w are in different simplices that have v as a vertex because its the only non trivial case (simplex is convex), then literally just draw a path to the v from both of them (again convex). the other cooler less trivial but more aura way is to show that its trivial for a point on the boundary, and that if its in the interior, then its connected and open and hence path connected (because i "proved" it on one of my 217 assignments). QED

3. (a) Show directly that the polyhedron of Example 1.3.7 is not locally compact. (b) Show that, in general, if a complex K is not locally finite, then the space $|K|$ is not locally compact

Solution 4.0.3. so this is basically the parabola and edge thing with intersections. its not locally compact because theres a point $(1, 0)$ that is not in that thing so that has to do with something, uhh my phone is going off rn dammit its this guy who sends me like 8000 reels a minute but its ok sometimes theyre funny gimme a sec.. nvm it was my sister asking if i watched hetalia (i havent). mannn im hungry imma go eat also i gotta review some topology quick

ive returned from work (we're so back). i should list the definitions of local compactness and local finiteness so that it becomes obvious for what im about to say. also, the point is $(0, 0)$, not $(1, 0)$ at all, because thats not even in the set so why would we talk about it...

local compactness: $\forall x \in K$ there is a neighbourhood U about x such that there is a compact C and $U \subset C$. now, a:

a) Every point in $|K|$ needs such a property, but we show that $(0, 0)$ does not. Suppose it did, then we could find some $U \subset |K|$ open in $|K|$ containing $\mathbf{0}$. U intersects with every single simplex σ_i with vertices $\mathbf{0}$ and $(1, \frac{1}{i})$. $U \cap \text{int } \sigma_i$ is open in $|K|$, and contains the first little bit of a line segment of each vertex. So, because its nonempty, we can choose a set

$A = \{x_i \in U \cap \text{int } \sigma_i \mid i \in \mathbb{N}\}$. Because $A \cap \sigma_i = \{x_i\}$, it is closed in the polyhedron topology. Now, A is a subspace that is closed, but $A \subset U \subset C$ and C exists and is compact by assumption. But A is a set that is infinite with discrete subspace topology: every subset $B \subset A$ intersects σ_i with $\{x_i\}$ which is closed, or not at all (and $\emptyset$ is still closed). $A = \bigcup_{i=1}^{\infty} \{x_i\}$, but there exists no finite subcover of this union. But, being a closed subset of a compact set, it must also be compact (because we are in a Hausdorff space). Thus, a contradiction.

b) Suppose K was not locally finite: we look at the 'bad' vertex v . There are infinitely many edges $[v, w_i]$. The same argument works. Suppose that the point v in $|K|$ was locally compact, then choose $U \subset C$ with U open in $|K|$ and C compact in $|K|$. Letting $A = \{x_i \in U \cap \text{int } \tau_i \mid i \in \mathbb{N}\}$, where τ_i is the simplex generated by v and w_i , we see that A is again an infinite set. Noting that every subset of A is closed because it intersects only one point of τ_i or nothing, A is again infinite discrete and hence not compact, which is a contradiction. QED

Exercises for Section 1.5: Free Abelian Groups

1. Show that if G is a finitely-generated Abelian group, then every subgroup $H < G$ is finitely-generated.

Solution 4.0.4. We can use that for such a G , there exists a surjective homomorphism $\varphi : \mathbb{Z}^n \rightarrow G$ that sends $(k_1, \dots, k_n) \mapsto \sum k_j g_j \in G$, for some n and some elements $\{g_1, \dots, g_n\} \subset G$. If $H < G$, we have that H is finitely-generated if and only if $\varphi^{-1}(H) < \mathbb{Z}^n$ is finitely-generated. Thus, it suffices to prove this statement for $\mathbb{Z}^n$.

The case for $n = 1$ is trivial, since every subgroup is of the form $k\mathbb{Z} < \mathbb{Z}$. Suppose that this holds for $\mathbb{Z}^{n-1}$. Let $h \in H < \mathbb{Z}^n$. The projection onto $n - 1$ coordinates is a surjective homomorphism $\pi : \mathbb{Z}^n \rightarrow \mathbb{Z}^{n-1}$, and $\ker \pi \cong \mathbb{Z}$. By assumption, $\pi(h) = \sum k_j \pi(h_j) \in \mathbb{Z}^{n-1}$ for some $h_1, \dots, h_{n-1} \in H$ (we can do this of course because π is surjective). Since $\pi(h) - \sum k_j \pi(h_j) = \pi(h - \sum k_j h_j) = 0$, we have that $h - \sum k_j h_j \in \ker \pi \cap H \cong H' < \mathbb{Z}$ for some subgroup H' of $\mathbb{Z}$, which is finitely-generated. Therefore h is generated by $h_1, \dots, h_{n-1}$ and a generator of $\ker \pi \cap H$. Since h was arbitrary, $H < \mathbb{Z}^n$ is finitely-generated.

2. (a) Show that if G is free, then G is torsion-free.

Solution 4.0.5. Let $\{g_\alpha\}_{\alpha \in \mathbb{J}}$ be a basis for G . Suppose $g = \sum n_i g_i$ was a torsion element of G , so that for some $m \in \mathbb{Z}$, $mg = \sum mn_\alpha g_\alpha = 0$. By uniqueness, this implies that $mn_\alpha = 0 \implies g = 0$ for all $\alpha \in \mathbb{J}$. Thus the torsion group is trivial.

- (b) Show that if G is finitely generated and torsion-free, then G is free.

Solution 4.0.6. By the fundamental theorem of finitely generated abelian groups, this is obvious. But because I am unconvinced that its meant to be this easy, and because I have already spent so long on this, I present another proof.

Solution 4.0.7. BEHOLD:

G is generated by a set of elements $\{g_1, \dots, g_n\}$ by assumption, and suppose without loss of generality, it is minimal. It suffices to show that this set is linearly independent. Suppose $a_1 g_1 + \dots + a_n g_n = 0$ for some $a_1, \dots, a_n \in \mathbb{Z}$. Because G is torsion-free, we can choose these numbers such that $\gcd\{a_1, \dots, a_n\} = 1$.

We essentially want to "collapse" the set of generators into a smaller set. Since the linear combination is equal to 0 by assumption, we want to show that if our generating set was not a linearly independent, then it is not minimal. We will set $y = a_1 g_1 + \dots + a_n g_n = 0$ and we will prove that we can force y to be a new generator (along with other generators $z_2, \dots, z_n$) that generate the same group, and that since $y = 0$, this new set of generators is strictly smaller, contradicting our choice of minimality. We demonstrate the procedure on only two generators and then reuse it inductively:

Let u, v be generators such that $y = au + bv$. Without loss of generality, let $\gcd\{a, b\} = 1$ so that $ra + sb = 1$ for some $r, s \in G$. Let $z = su - rv$. Thus $ry = r(au + bv) = rau + rbv = (1 - sb)u + rbv = u - sbu + rbv \implies ry - bz = u$. Similarly, $v = sy - az = s(au + bv) - az = sau + sbv - az = sau + (1 - ra)v - az = sau + v - rav - asu + arv = v$. Both u and v can be recovered, and hence these new generators y, z generate the same group.

What we have shown is that by using invertible transformations under the integers, we are able to generate the same group by forcing $y = au + bv$ to be one of the generators. Applying this same technique to our problem at hand, we let $y = \sum a_i g_i$, noting for the future that $y = 0$. We define $d_k = \gcd\{a_1, \dots, a_k\}$ so that $d_{k+1} \mid d_k$

(and allowing $d_1 = a_1$). Thus, we can use the trick above to coalesce $y = a_1g_1 + a_2g_2 + a_3g_3 + \dots$ into $y = d_2p_1 + a_3g_3 + \dots$, where $p_1 = \frac{a_1}{d_2}g_1 + \frac{a_2}{d_2}g_2$ (note that these are integer coefficients still). This gives us a new generator z_1 that we keep track of for now. Continuing, we obtain $p_2 = \frac{d_2}{d_3}p_1 + \frac{a_3}{d_2}g_3$ to get $y = d_3p_2 + \dots$ and a new generator z_2 . Continuing all the way, we eventually obtain $y = d_np_{n-1} = 0$ and generators $z_1, \dots, z_{n-1}$, which all generate the same group G . Since $y = d_np_{n-1} = 0$ (and again, G is torsion-free), we found a new set of generators $\{z_1, \dots, z_{n-1}\}$ strictly smaller than our original set $g_1, \dots, g_n$, contradicting minimality. Therefore, $a_1 = a_2 = \dots = a_n = 0$ and $\{g_1, \dots, g_n\}$ is a basis, thus G is free.

- (c) Show that the additive group of rationals $\mathbb{Q}$ is torsion free but not free. [Hint: If $\{g_\alpha\}$ is a basis for $\mathbb{Q}$, let β be fixed and express $g_\beta/2$ in terms of this basis.]

Solution 4.0.8. This one is a lot easier than the previous one. $\mathbb{Q}$ is obviously torsion-free because it's a field. If the given basis was indeed a basis, then we could represent $g_\beta/2 = \sum a_\alpha g_\alpha$. Multiplying by 2, we re-obtain $g_\beta = \sum 2a_\alpha g_\alpha$. By uniqueness, our coefficients being odd on the LHS and even on the RHS implies that the set is not a basis.

3. (a) Let $\gcd(m, n) = 1$. Show that $\mathbb{Z}/m \oplus \mathbb{Z}/n \cong \mathbb{Z}/mn$:

Solution 4.0.9. Given $(1_m, 1_n) \in \mathbb{Z}/m \oplus \mathbb{Z}/n$, we have that $(1_m, 1_n)^k = (1_m^k, 1_n^k) = (0, 0) \iff m \mid k \text{ and } n \mid k$. $k = mn$ is the lowest common multiple of m and n and $(1, 0), (0, 1)$ generates $\mathbb{Z}/m, \mathbb{Z}/n$ respectively. This suffices the claim.

- (b) $G \cong \mathbb{Z}/18 \oplus \mathbb{Z}/36$. Express G as a direct sum of cyclic groups of prime power order:

Solution 4.0.10. Since $18 = 2 \cdot 3^2$ and $36 = 2^2 \cdot 3^2$, we get after grouping (pun intended) together the primes, that $G \cong \mathbb{Z}/2 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/9 \oplus \mathbb{Z}/9$.

- (c) $G \cong \mathbb{Z}/2 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/9$; Find the torsion coefficients of G :

Solution 4.0.11. By the Chinese Remainder Theorem, we can just group together the primes in increasing order to get the torsion elements. We pad by 1 the first prime (2 in this case) to get it up to

three elements. so we have $1 \mid 2 \mid 4$ and $3 \mid 3 \mid 9$. Multiplying, we get $1 \cdot 3 \mid 2 \cdot 3 \mid 4 \cdot 9$ giving us 3, 6, 36 as our torsion coefficients. Of course, the padding-by-1 trick is valid by what we proved in (a), taking $n = 1$ and $m = 3$. We essentially obtain $G \cong \mathbb{Z}/3 \oplus \mathbb{Z}/6 \oplus \mathbb{Z}/36$.

(d) $G \cong \mathbb{Z}/15 \oplus \mathbb{Z}/20 \oplus \mathbb{Z}/18$; Find the torsion coefficients of G :

Solution 4.0.12. Decomposing into its primes, we get 3, 5 and 4, 5 and 2, 9. Using CRT and grouping the primes, we get $2 \mid 4, 3 \mid 9, 5 \mid 5$, so $2 \cdot 3 \cdot 5 \mid 4 \cdot 9 \cdot 5$ which gives $30 \mid 180$ as our torsion coefficients. So, $G \cong \mathbb{Z}/30 \oplus \mathbb{Z}/180$.

Exercises for Section 1.6: Homology Groups

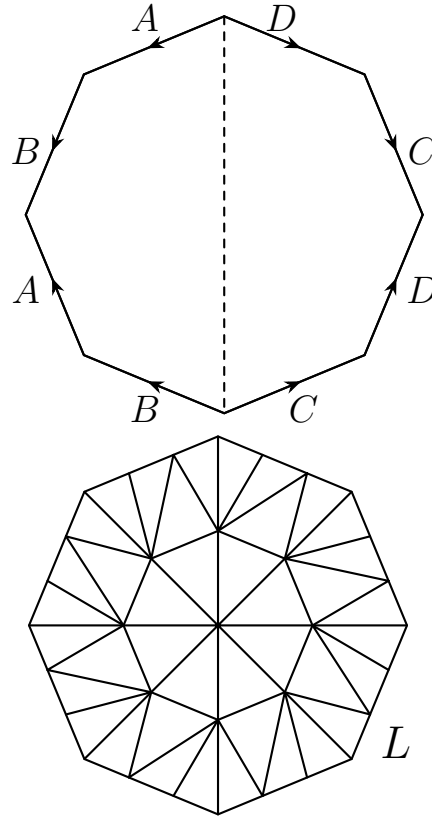
1. i did 1-4 in my notebook (its light work, the last two are actually hard).
– this is also where we'll actually need to learn how to put pictures lol
- 2.
- 3.
- 4.

Exercises for Section 1.7: Homology of Surfaces

1. Let w_1 and z_1 be the cycles on the Klein bottle as in 3.4. Show that $w_1 + z_1$ represents a generator of the infinite cyclic group $H_1(S)/T_1(S)$.

Solution 4.0.13. By the theorem, $w_1 + T_1(S)$ generates The free subgroup (isomorphic to) $H_1(S)/T_1(S)$. In particular, since $w_1 + T_1(S) = w_1 + z_1 + T_1(S)$, we have that $\langle w_1 \rangle + T_1(S) = \langle w_1 + z_1 \rangle + T_1(S) \cong H_1(S)/T_1(S)$.

2. The connected sum $T \# T$ of two tori can be represented as a quotient space of an octagonal region in the plane by making identifications on the boundary as indicated in Figure 1.7.12. [Splitting this octagon along the dotted line gives two tori with open discs deleted.]
- 3.
- 4.
- 5.



Exercises for Section 1.8: 0-Dimensional Homology

1. Let G be an abelian group and let $\phi : G \rightarrow \mathbb{Z}$ be an epimorphism. Show that G has an infinite cyclic subgroup H such that $G = (\ker \phi) \oplus H$. [Hint: Define a homomorphism $\psi : \mathbb{Z} \rightarrow G$ such that $\phi \circ \psi$ is the identity; let $H = \text{im } \psi$.]

Solution 4.0.14. Let $I = \phi^{-1}\{1\}$ and let $1_G \in I$ be a representative. We must have that $\phi(1_G) = 1$, $\phi(1_G + 1_G) = 2$, and so on. By induction $\phi(1_G^n) = \phi(1_G)^n = n$. We may define $\psi : \mathbb{Z} \rightarrow G$ such that $\psi(n) = (1_G)^n$, and this is well defined: for if $\tilde{1}_G \in I$ was different than 1_G , then $\phi(\tilde{1}_G^n) = \phi(\tilde{1}_G)^n = n = \phi(1_G^n)$. Since ψ is injective, we have that $H := \psi(\mathbb{Z}) \leq G$ is an infinite cyclic subgroup of G , and $\phi \circ \psi(n) = \phi(1_G^n) = n$, so $\phi \circ \psi = \text{id}_{\mathbb{Z}}$. The kernel of ϕ may have infinite elements, but since it is the identity when composed with ψ , we have that $\ker \phi \cap \text{im } \psi = \{0\}$,

so they are direct summands of G . That is,

$$G \cong \ker \phi \oplus H$$

2. Show that if $\phi : C_0 \rightarrow Z$ is any epimorphism such that $\phi \circ \partial_1 = 0$, then $H_0(K) \cong (\ker \phi) / (\text{im } \partial_1) \oplus \mathbb{Z}$.

Solution 4.0.15. By (a), $C_0 \cong \ker \phi \oplus H$, so

$$H_0(K) \cong C_0 / \text{im } \partial_1 \cong \ker \phi / \text{im } \partial_1 \oplus H \cong \ker \phi / \text{im } \partial_1 \oplus \mathbb{Z}$$

Exercises for Section 1.9: Homology of a Cone

1. Let K be a complex; let $w_0 * K$ and $w_1 * K$ be two cones on K whose polytopes intersect only in $|K|$.

- (a) Show that $(w_0 * K) \cup (w_1 * K)$ is a complex.

Solution 4.0.16. We must verify that every face of a simplex in $L = (w_0 * K) \cup (w_1 * K)$ is also itself in L . We also verify that a non-empty intersection between simplices σ and τ is a face of each of them. Indeed, if some simplex ρ lies in L , then since $|w_0 * K| \cap |w_1 * K| = |K|$, ρ either lies entirely in one of $|w_i * K|$, or in $|K|$. Since these are all complexes, every face of ρ lies in the same complex and hence lies in L . Similarly, the intersection of σ and τ lies entirely in one of these complexes and hence must be a face of both of them.

- (b) Using the bracket notation, define $\phi : C_p(K) \rightarrow C_{p+1}(S(K))$ by the equation

$$\phi(c_p) = [w_0, c_p] - [w_1, c_p]$$

Show that ϕ induces a homomorphism

$$\phi_*(c_p) : \tilde{H}_p(K) \rightarrow \tilde{H}_{p+1}(S(K))$$

Solution 4.0.17. It suffices to show that ϕ sends boundaries to boundaries and cycles to cycles, because for the coset $\gamma = z_p + B_p(K)$, we would then have $\phi(\gamma) = \phi(z_p) + B_{p+1}(S(K)) \in \tilde{H}_{p+1}(S(K))$. Now, for an arbitrary $c \in C_p(K)$, we have that

$$\partial \circ \phi(c) = \partial[w_0, c] - \partial[w_1, c] = c - c - [w_0, \partial c] + [w_1, \partial c] = -\phi \circ \partial(c).$$

Thus we easily see that if $\partial(c) = 0$ then $\partial \circ \phi(c) = 0$ and if $c = \partial d$ then $\phi(c) = \partial(-\phi(d))$, which suffices the claim.

- (c) Show that ϕ_* is an isomorphism when K consists of the proper faces of a 2-simplex. We in fact have (but will not prove) that ϕ_* is an isomorphism in general.

Solution 4.0.18. Roughly speaking, the suspension is obtained by "capping" the top and bottom of the hollow triangle in 3-d space, and is thus homeomorphic to $\mathbb{S}^2$. Its homology groups are thus $\tilde{H}_0(S(K)) \cong \tilde{H}_1(S(K)) \cong \{0\}$, and $\tilde{H}_2(S(K)) \cong \mathbb{Z}$.

Clearly $\tilde{H}_0(K) \cong \{0\}$ which is isomorphic to $\tilde{H}_1(S(K))$. This is also clear from the fact that ϕ_* is a homomorphism of groups and can only send $0 \mapsto 0$.

We have that $\tilde{H}_1(K)$ is generated by $z = [v_0, v_1] \oplus [v_1, v_2] \oplus [v_2, v_0]$. Thus $\phi_*(z) \in Z_2(S(K))$ sends z to its corresponding homology class. Since torsion is preserved under homomorphism, and since the rank of $\tilde{H}_2(S(K))$ is 1, we have that $\phi_*(z)$ generates $\tilde{H}_2(S(K))$, which suffices the claim. It is clear that all other values of p result in trivial and hence isomorphic homology groups between both complexes.